Decentralised Bandits on a Graph

No server: agents talk only to neighbours

CS6007 project notes

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What this folder is about. Agents sit on a communication graph and exchange information with neighbours. The graph enters regret through quantities such as the diameter, spectral gap, clique-cover or independence numbers. Two protocol families appear: *relaying* exact information (Kolla et al., Dubey & Pentland) and *gossip / consensus averaging* (Chawla et al., Amani & Thrampoulidis). Our Paper A lives here.

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1 Kolla, Jagannathan & Gopalan (2016/2018): bandits over a social network

R. K. Kolla, K. Jagannathan, A. Gopalan. Collaborative Learning of Stochastic Bandits over a Social Network. IEEE/ACM Trans. Networking 2018 (arXiv:1602.08886). File: `1602.08886v2.pdf`

Setting.

- Agents on a social network play the same stochastic MAB; each reward is seen instantly by the agent *and its neighbours* (one-hop sharing).

What they did.

- Analyses **UCB-Network** (every agent runs UCB on all observed rewards).
- Defines non-altruistic, individually consistent (NAIC) policies and lower-bounds their regret.
- Proposes a **dominating-set leader–follower** policy that exploits the network structure.

Main results.

- An upper bound for UCB-Network with a graph-dependent constant (a combinatorial optimisation), specialised to complete and star graphs.
- A universal network-independent lower bound for reasonable policies, and a graph-dependent one.
- Natural extensions of single-agent policies can perform badly; leader–follower policies on dominating sets do much better.

Why it matters for our project.

- An early proof that "just run the single-agent algorithm on shared data" is not enough on a graph.

Open problems.

- **Our reading:** Sharing is one-hop and instantaneous: no relaying, no delays, no communication budget.
- **Our reading:** Gaps remain between the upper and lower bounds on general graphs.

2 Chawla, Sankararaman, Ganesh & Shakkottai (AISTATS 2020): gossiping insert-eliminate

R. Chawla, A. Sankararaman, A. Ganesh, S. Shakkottai. The Gossiping Insert-Eliminate Algorithm for Multi-Agent Bandits. AISTATS 2020 (arXiv:2001.05452).
File: `2001.05452v4.pdf`

Setting.

- N agents on an arbitrary connected graph solve the same MAB; pairwise gossip with a communication budget.

What they did.

- Each agent plays only a small *subset* of arms; agents gossip only **arm IDs** (recommendations), not samples, and update their subsets (insert/eliminate).

Main results.

- If agents communicate $\Omega(\log T)$ times, every agent's regret is about N times smaller than without collaboration.
- Communication constraints only affect a second-order term, with explicit regret–communication trade-off bounds.

Why it matters for our project.

- A clear example of redesigning *what* is communicated (IDs, $O(\log K)$ bits) rather than federating an existing algorithm.
- Suggests a "recommend-then-pull" variant for bins in our Phase 2.

Open problems.

- **Our reading:** Relies on all agents sharing the same best arm; heterogeneous agents or malicious recommenders break it (see Vial et al., folder 05).
- **Our reading:** Multi-armed only; contextual or structured arms are not covered.

3 Amani & Thrampoulidis (AAAI 2021): decentralised linear bandits (DLUCB)

S. Amani, C. Thrampoulidis. Decentralized Multi-Agent Linear Bandits with Safety Constraints. AAAI 2021 (arXiv:2012.00314).
File: `arxiv-2012.00314.pdf`

Setting.

- N agents on an arbitrary fixed graph; linear bandit; gossip through a doubly stochastic communication matrix; no server.

What they did.

- **DLUCB**: UCB plus accelerated (Chebyshev) consensus run in cycles.
- **RC-DLUCB**: rare communication.
- **Safe-DLUCB**: linear safety constraints.

Main results.

- Regret $\tilde{O}(d\log(NT)\sqrt{NT})$, the centralised rate, plus a small additive "**regret of delay**" that depends on the spectral gap.
- Communication $O(dN^2)$ per round for DLUCB; RC-DLUCB trades a little regret for much less communication.

Why it matters for our project.

- The template for tightening our delay term: their delay cost is additive and only logarithmic, while ours is $N \cdot D \cdot N_{\text{bins}}$.
- One of the protocols we compare against (Chebyshev gossip in E14–E20).

Open problems.

- **Stated by the authors:** Decentralised algorithms with guarantees on **time-varying networks**.
- **Stated by the authors:** Extensions to **nonlinear settings**.
- **Stated by the authors:** Other types of safety constraints.

4 Dubey & Pentland (ICML 2020): cooperative kernel bandits (Coop-KernelUCB)

A. Dubey, A. Pentland. Kernel Methods for Cooperative Multi-Agent Contextual Bandits. ICML 2020 (arXiv:2008.06220).
File: arxiv-2008.06220.pdf

Setting.

- Agents on a graph with delays; kernelised contextual bandits; agents may have different but related reward functions, linked by a network kernel.

What they did.

- **Coop-KernelUCB**: message passing (the LOCAL protocol) where observations are forwarded hop by hop with a time-to-live γ; product kernels for heterogeneous agents.
- A variant, EAGER-KernelUCB, that performs better in experiments but has no proof.

Main results.

- Average pseudo-regret $\tilde{O}(\sqrt{\bar{\chi}(G_\gamma)T/V})$, where $\bar{\chi}(G_\gamma)$ is a clique-cover number of the γ-th graph power.
- Outperforms single-agent and naive cooperative baselines on synthetic and real networks.

Why it matters for our project.

- The closest prior work to Paper A: distributed, unknown reward, a graph, and a *relay* protocol. It argues, as we found, that averaging is biased and message passing is preferable.
- Our main candidate for a baseline that can learn an unknown non-linear reward.

Open problems.

- **Stated by the authors:** **Lower bounds** on group regret in cooperative decision-making with delays (the limits of cooperation).
- **Stated by the authors:** Prove a guarantee for EAGER-KernelUCB (conjectured $\tilde{O}(\sqrt{(\gamma+\alpha(G_\gamma))T/V})$ in the linear case).
- **Stated by the authors:** Extend to the Bayesian case.
- **Our reading:** Communication grows linearly with T (raw observations every round); event-triggered relaying is not studied.

5 Open problems across this folder

- **Graph-dependent lower bounds** are open (Dubey & Pentland state this).
- **Time-varying graphs** and **nonlinear rewards** are open for gossip methods (Amani & Thrampoulidis state this).
- Delay costs appear in very different forms: additive with the spectral gap (DLUCB), a clique cover under a square root (Coop-Kernel). No paper gives a tight additive delay term for non-parametric or sleeping bandits.

- Nothing here combines an unknown link with an index direction on a graph: the gap our project fills.